

BUSINESS PROBLEM STATEMENT

A financial services organization seeks to strengthen its ability to detect and prevent fraudulent financial transactions in order to reduce financial losses, improve transaction security, and enhance customer trust in digital payment systems. The increasing growth of online banking, electronic transfers, and digital payment platforms has resulted in a rise in fraudulent activities, making it difficult for traditional fraud detection methods to identify suspicious transactions accurately and efficiently.

Management aims to gain deeper insights into transaction behavior and identify the key factors associated with fraudulent activities, including transaction type, transaction amount, sender and receiver account balances, and unusual transaction patterns. By leveraging machine learning and transaction data analysis, the organization intends to develop a predictive fraud detection system capable of identifying high-risk transactions in real time.

The primary business objective for this project is to answer the following question:

“How can machine learning and transaction data be utilized to detect fraudulent financial transactions, improve transaction security, and support faster fraud prevention decision-making?”

Project Deliverables

1. Data Preparation and Modeling (Python)

Clean, preprocess, and transform raw transaction data into a structured format suitable for machine learning analysis and fraud prediction. Build and train a machine learning classification model capable of identifying fraudulent transactions.

2. Data Analysis and Machine Learning (Python)

Perform exploratory data analysis to identify fraud trends, transaction behavior patterns, and relationships between transaction features. Evaluate model performance using classification metrics to determine prediction accuracy and reliability.

3. Fraud Detection Application (Streamlit)

Develop an interactive fraud detection application that allows users to input transaction details and receive real-time fraud predictions using the trained machine learning pipeline.

4. GitHub Repository

Maintain a well-structured repository containing all project components, including Python scripts, machine learning models, datasets, application files, documentation, and deployment resources.